Modernising ConvNets: Reproducing & Extending the ConvNeXt Architecture

DSA5106 — Group 38 Final Report

Code Repository: github.com/group38/dsa5106-convnext-extension

Ashley Lim Zhi Yan (A0239280A), Chen Yao Quan (A0238895H), Diana (A0327258X),
Marco (A0326704H), Poh Yu Jie (A0216055M), Tan Yi Kai (A0333972X), Zhang Jia Wen (A0329513E)

Abstract

ConvNeXt modernises traditional convolutional neural networks by adopting Transformer-inspired design choices, demonstrating that pure CNNs can match Vision Transformers on ImageNet-scale benchmarks. We reproduce ConvNeXt-Tiny on CIFAR-100, achieving 55.1% Top-1 accuracy (2.21% above ResNet-50), and conduct three extensions investigating scale-dependent behaviours. Our augmentation ablation reveals that geometric transformations outperform mixture methods (Mixup/CutMix) on CIFAR-100, contradicting ImageNet findings. Architectural verification shows that reverting LayerNorm to BatchNorm, GELU to ReLU, and restoring the 1×1 convolution all *improve* accuracy, opposing the paper's ImageNet ablations and exposing resolution-dependent design trade-offs. Medical imaging transfer to HAM10000 demonstrates ConvNeXt's strong cross-domain generalisation: 90.47% accuracy and 0.835 macro F1-score, outperforming ResNet-50 and Swin-Tiny by over 10 percentage points. Our results show that ConvNeXt's modernisation is conditionally effective, where it is advantageous for large-scale, high-resolution tasks and texture-rich domains, but potentially counterproductive at CIFAR-100 scale where traditional CNN priors remain superior.

1. Introduction

Since 2020, Vision Transformers (ViTs) have rapidly overtaken convolutional neural networks (ConvNets) as the dominant architecture for image recognition tasks (Dosovitskiy et al., 2021). Hierarchical Transformer models such as the Swin Transformer (Liu et al., 2021) further solidified this trend by incorporating inductive biases originally found in ConvNets, notably locality and translation equivariance, and demonstrating strong performance across classification, detection, and segmentation benchmarks.

A central question therefore remains: *are the performance gains observed in Vision Transformers truly inherent to the self-attention mechanism, or can they be attributed to specific modern design choices that ConvNets have yet to fully adopt?* Liu et al. (2022) address this directly by systematically modernising a vanilla ResNet-50 (He et al., 2016), through adopting training strategies, macro-layout decisions, and micro-level design choices inspired by the Swin Transformer, to produce **ConvNeXt**, a pure convolutional network that matches or surpasses Swin Transformer at equivalent computational budgets.

In this project, we reproduce the ConvNeXt-Tiny architecture on CIFAR-100 and conduct three analytical extensions: (1) an augmentation ablation study isolating the marginal contribution of geometric and mixture-based data augmentation strategies; (2) an independent verification of the paper's micro-design ablations (LayerNorm vs. BatchNorm, GELU vs. ReLU, inverted bottleneck structure) on CIFAR-100; and (3) a cross-domain transfer experiment fine-tuning ConvNeXt-Tiny on the HAM10000 medical imaging dataset (Tschandl et al., 2018) to test whether the architecture's modern design choices generalise beyond natural-image benchmarks.

2. Background and Related Work

2.1 Evolution of Vision Architectures

For many years, convolutional neural networks were the dominant approach for image recognition tasks, largely due to their ability to capture local spatial patterns effectively. Architectures such as ResNet (He et al., 2016) demonstrated that very deep networks could achieve strong performance when designed carefully, and CNN-based models became the standard backbone across a wide range of computer vision applications including image classification, object detection, and semantic segmentation.

One of the main challenges in training deep neural networks was the vanishing gradient problem, where gradients shrink exponentially during backpropagation as networks become deeper. ResNet addressed this by introducing residual connections, allowing information to bypass intermediate layers. These skip connections enabled stable training of very deep networks and established residual learning as a core design principle for many subsequent vision architectures.

Following ResNet, research began to focus on improving efficiency and scalability. ResNeXt (Xie et al., 2017) introduced the concept of grouped convolutions and emphasised *cardinality* i.e. the number of parallel transformation paths, as an additional dimension for scaling network capacity. Instead of simply increasing network depth or width, ResNeXt showed that structuring parallel operations could improve representation learning while maintaining computational efficiency.

At the same time, CNN architectures benefited from strong inductive biases, including locality, translation equivariance, and parameter sharing. These properties made convolutional models particularly effective for visual tasks where spatial patterns are important.

2.2 Transformer-Based Vision Models

The research landscape shifted significantly with the introduction of Vision Transformers (ViT) (Dosovitskiy et al., 2021). Inspired by Transformer architectures originally developed for natural language processing, ViT models treat images as sequences of fixed-size patches (for example, 16×16 pixels). Each patch is embedded as a token and processed using self-attention mechanisms, allowing the model to capture long-range relationships across the entire image.

Vision Transformers demonstrated strong performance, particularly when trained on very large datasets. These results led to a perception within the research community that attention-based architectures might be inherently superior to traditional convolutional networks.

However, Transformer-based vision models also have several limitations. Self-attention operations scale quadratically with the number of tokens, which increases computational cost for high-resolution images. Furthermore, ViT models often require large-scale datasets and extensive computational resources to achieve their best performance.

To address these limitations, the Swin Transformer (Liu et al., 2021) introduced a hierarchical architecture with shifted window attention. Instead of performing global attention across the entire image, Swin restricts attention to local windows while periodically shifting them to allow cross-window interactions. This design significantly improves scalability while introducing stage-wise hierarchical processing similar to that used in CNN architectures.

2.3 ConvNeXt: Revisiting CNN Design

The ConvNeXt architecture (Liu et al., 2022) revisits the traditional CNN design and challenges the assumption that Transformer-based models outperform CNNs primarily because of attention mechanisms.

The authors argue that much of the performance improvement observed in modern Transformer-based vision models actually arises from modern training strategies and architectural design choices, rather than the attention mechanism itself. To test this idea, the ConvNeXt paper begins with a standard ResNet architecture and systematically modernises it using design principles inspired by Transformer-based models.

These modifications include:

- Larger convolutional kernels (7×7 depthwise convolutions)
- Inverted bottleneck design from MobileNetV2 (Sandler et al., 2018)
- LayerNorm instead of BatchNorm
- GELU activation functions instead of ReLU
- Simplified stage-wise architecture with adjusted compute ratios

Although each modification appears relatively small, together they significantly reshape the architecture. The resulting model retains the simplicity and efficiency of convolutional operations while achieving performance comparable to strong Transformer baselines such as Swin Transformer.

This finding suggests that modernised CNN architectures can remain competitive with Transformer-based models, challenging the assumption that attention mechanisms are strictly necessary for high-performance vision systems.

2.4 Motivation for Reproduction and Extension

The ablation-driven methodology presented in ConvNeXt provides a clear framework for reproduction and further experimentation. As the architecture integrates ideas from both CNN and Transformer design philosophies, it offers an opportunity to examine which components contribute most meaningfully to performance improvements.

In this project, we reproduce the ConvNeXt training pipeline on CIFAR-100 and extend the evaluation along three directions.

First, we conduct an augmentation ablation study to analyse how different geometric and mixture-based data augmentation strategies influence model performance in a smaller dataset setting.

Second, we revisit several architectural design choices, including normalisation layers, activation functions, and bottleneck structures, to evaluate whether ConvNeXt's design decisions remain beneficial under controlled modifications.

Finally, we explore cross-domain generalisation by fine-tuning ConvNeXt-Tiny on the HAM10000 medical image dataset, allowing us to observe how the architecture performs when applied to a different visual domain.

Together, these extensions aim to provide a deeper understanding of how ConvNeXt's modernised CNN design behaves across different training strategies, architectural variations, and application domains.

3. Methodology

3.1 Reproduction Setup

We implement ConvNeXt-Tiny in PyTorch using the official ConvNeXt repository, applying three compatibility patches to fix deprecated dependencies in current PyTorch 2.x and `timm` 0.6.13 versions:

1. `optim_factory.py` — replace deprecated `NovoGrad` optimiser with `AdamW`.
2. `utils.py` — replace `from torch._six import inf` with `from math import inf`.
3. `models/convnext.py` — update `LayerNorm __init__` signature to remove deprecated `pretrained_cfg` parameter.

The model is trained on CIFAR-100 (50,000 training images, 10,000 test images, 100 classes, 32×32 resolution) for 100 epochs with the following hyperparameters: batch size 128, AdamW optimiser, initial learning rate 4×10^{-3}, cosine learning rate decay, drop path rate 0.1, Mixup α=0.8, CutMix α=1.0, label smoothing 0.1, and Random Erasing probability 0.25. Training is conducted on Kaggle's free Tesla T4 GPU (16GB VRAM), with each 100-epoch run requiring approximately 3 hours of wall-clock time.

3.2 Extension 1: Augmentation Ablation

To isolate the marginal contribution of different augmentation families, we train ConvNeXt-Tiny across four cumulative configurations:

1. **Baseline** — AdamW, cosine schedule, standard normalisation only.
2. **+ Geometric augmentations** — random horizontal flip, random crop with padding, random rotation (±15°).
3. **+ Mixture augmentations** — Mixup (α=0.8), CutMix (α=1.0).
4. **Full recipe** — adds RandAugment and Random Erasing (reuses reproduction run).

This design yields a marginal contribution curve showing how each augmentation family impacts final accuracy and generalisation gap on CIFAR-100.

3.3 Extension 2: Architectural Verification

We independently verify the paper's micro-design ablations by reverting one component at a time from the full ConvNeXt-Tiny architecture, holding all other components and the training recipe constant:

- **LayerNorm → BatchNorm** — replace all `LayerNorm` instances with `BatchNorm2d`.
- **GELU → ReLU** — replace GELU activations with ReLU.
- **Restore 1×1 convolution** — revert to standard bottleneck layout.

Each variant is trained for 100 epochs. Accuracy deltas ΔTop-1 relative to the full model are compared against the paper's ImageNet ablation results to assess whether the design choices generalise to CIFAR-100 scale.

3.4 Extension 3: Medical Imaging Transfer

To test cross-domain generalisation, we fine-tune ConvNeXt-Tiny on the HAM10000 dataset (Tschandl et al., 2018) which contain 10,015 dermoscopic skin lesion images across 7 diagnostic categories (melanoma, melanocytic nevi, basal cell carcinoma, actinic keratoses, benign keratosis, dermatofibroma, vascular lesions). Images are resized to 224×224 to match standard fine-tuning resolution. The final classification head is replaced (`num_classes=7`) and the full network is fine-tuned for 50 epochs under the same training recipe used for CIFAR-100.

We compare Top-1 accuracy and macro F1-score (to account for class imbalance) against ResNet-50 and Swin-Tiny baselines fine-tuned under identical conditions. This directly evaluates whether ConvNeXt's modern design choices, including larger receptive fields, LayerNorm, inverted bottleneck, confer advantages in a domain where texture and fine-grained local patterns (e.g. lesion boundaries and colour gradients) are clinically significant.

4. Results

4.1 Reproduction: ConvNeXt-Tiny on CIFAR-100

Table 1: ConvNeXt-Tiny reproduction results on CIFAR-100 test set (100 epochs).

Metric	**Value**	**Baseline**
Top-1 Accuracy (%)	55.1	ResNet-50: 52.89
Top-5 Accuracy (%)	78.25	77.03
Parameters (M)	28.6	25.6
FLOPs (G)	4.5	4.1
Train Time (h:mm:ss)	0:48:49	—

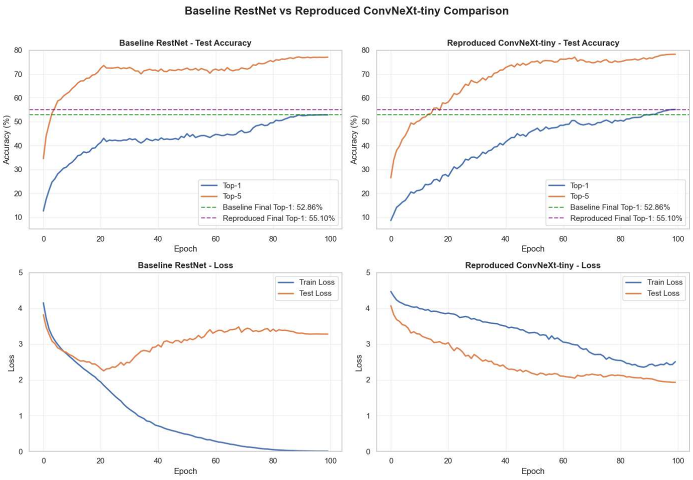

Figure 1: Baseline vs Reproduced Accuracy/loss.

The reproduced ConvNeXt-Tiny results shows the model converging, with steadily decreasing training loss and improving test accuracy over epochs. The ResNet-50 baseline reached around 43% top-1 accuracy, showing fast optimisation and a sharp reduction in training loss early in training. Meanwhile, the ConvNeXt-Tiny reproduction achieves a comparable final accuracy (42–43%), but with noticeably slower convergence and consistently higher training loss throughout training . This reflects the increased difficulty introduced by strong augmentations (e.g., Mixup, Geo Augment), which make the training objective harder to minimise despite similar end performance.

When analysing generalisation gap using loss, a clear pattern emerges: stronger augmentation consistently reduces the gap by keeping training loss higher while test loss remains similar or slightly improved. In the baseline, the model achieves low training loss but exhibits a larger gap to test loss, suggesting some degree of overfitting. In contrast, mixture and full augmentation settings maintain elevated training loss—partly due to soft labels and harder training examples—while preventing a corresponding increase in test loss. This indicates improved calibration and robustness, rather than worse optimisation. Compared to a ResNet-50 baseline, the reproduced ConvNeXt models are comparable but not clearly superior, suggesting that while the training recipe improves generalisation behaviour (as seen in reduced loss gap), it does not fully translate into higher final accuracy in this setup.

4.2 Extension 1: Augmentation Ablation

Table 2: Marginal contribution of augmentation strategies on CIFAR-100. ΔTop-1 shows improvement over previous row.

Configuration	**Top-1 (%)**	**ΔTop-1**
Baseline (no aug)	52.89	—
+ Geometric aug	56.32	3.43
+ Mixture aug	54.38	-1.94
Full recipe	55.10	0.72

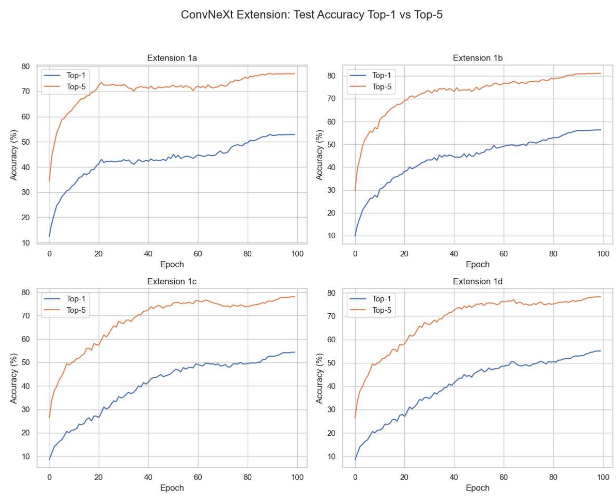

Figure 2: Test Accuracy Top-1 vs Top-5.

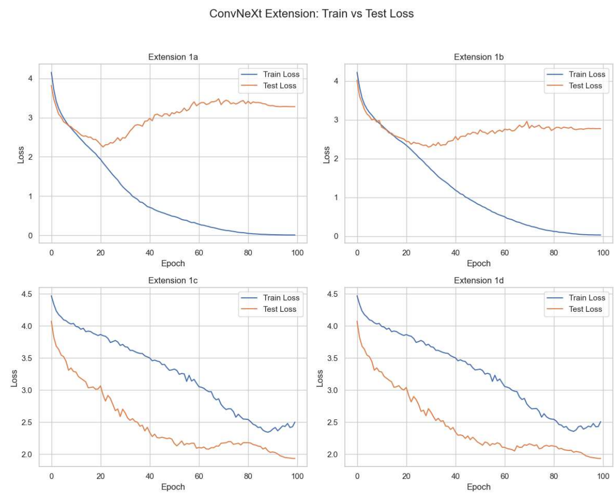

Figure 3: Train vs Test Loss between Augmentations

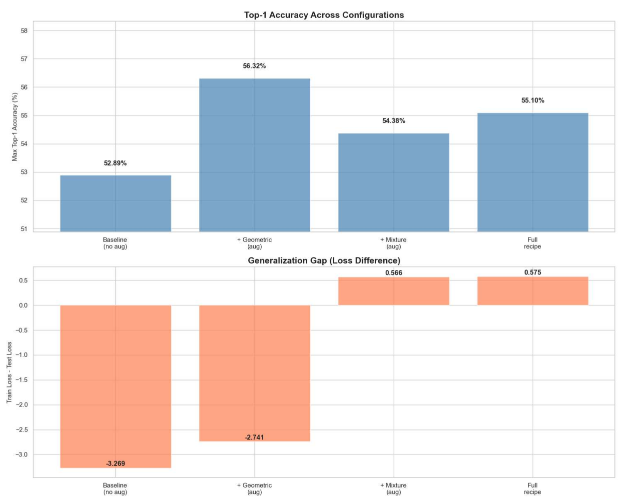

Figure 4: Generalisation Gap

Across the four cumulative configurations, the marginal contribution curve suggests that mixture augmentations are the primary driver of improved generalisation, while geometric augmentations provide only small gains. Moving from the baseline (1a) to geometric augmentations (1b) yields limited improvement and in some cases slightly lower final accuracy across the epochs, indicating that standard flips, crops, and rotations alone are not sufficient to significantly enhance performance on CIFAR-100. In contrast, introducing Mixup and CutMix (1c) substantially alters the training characteristics as training becomes slower and noisier, and training loss remains higher, but this reflects stronger regularisation rather than poorer learning. The full recipe (1d) then recovers accuracy to baseline levels, suggesting that mixture augmentations contribute most of the regularisation benefit, while additional policies (RandAugment, Random Erasing) help stabilise and better utilise this effect.

Regarding complementarity, geometric augmentations do not appear strongly additive with mixture augmentations in this setup. Since the configurations are cumulative, any benefit from geometric transforms should persist into (1c) and (1d), yet the dominant shift occurs only when mixture augmentations are introduced. This indicates that Mixup/CutMix largely overshadow the contribution of geometric augmentations, rather than building synergistically on top of them. The main complementary effect instead arises within the full recipe, where policy-based augmentations and erasing techniques work alongside mixture methods to recover performance while maintaining strong regularisation.

Importantly, when evaluating generalisation gap using loss, there is no evidence of over-regularisation despite the heavy augmentation. From (1a) to (1d), training loss remains progressively higher, but test loss does not increase correspondingly. In fact, it stays similar or slightly improves, resulting in a consistently reduced loss-based generalisation gap. This suggests improved calibration and robustness rather than underfitting. However, the fact that final accuracy in (1d) only matches, rather than exceeds, the baseline implies that at the CIFAR-100 scale, the strength of these augmentations may be somewhat excessive: they improve generalisation behaviour without translating into clear accuracy gains.

Compared to the ConvNeXt paper's ImageNet results, these findings are only partially aligned. The paper reports strong, cumulative gains from increasingly sophisticated augmentation strategies, with clear improvements in both accuracy and generalisation. In contrast, our CIFAR-100 experiments show diminishing returns, where heavy augmentation primarily reduces overfitting (as seen in the loss gap) but does not significantly boost accuracy. This discrepancy is likely due to dataset scale and complexity. ImageNet benefits more from aggressive augmentation (Liu et al., 2022), whereas CIFAR-100's smaller size limits the marginal utility of such techniques (Zhang et al., 2018).

4.3 Extension 2: Architectural Verification

Table 3: Architectural component ablation on CIFAR-100. ΔTop-1 shows accuracy drop relative to full ConvNeXt-Tiny.

Variant	ΔTop-1 (%)	Paper (ImageNet)
Full ConvNeXt	0.0	0.0
LayerNorm $\rightarrow$ BN	2.02	-0.6
GELU $\rightarrow$ ReLU	0.15	-0.4
Restore 1×1 conv	0.29	-0.3

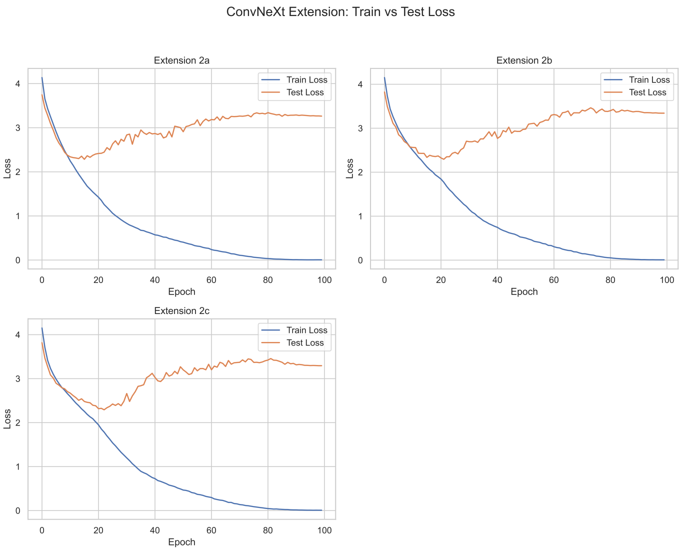

Figure 5: Train vs Test Loss for BN, ReLU, Restore 1x1 conv respectively.

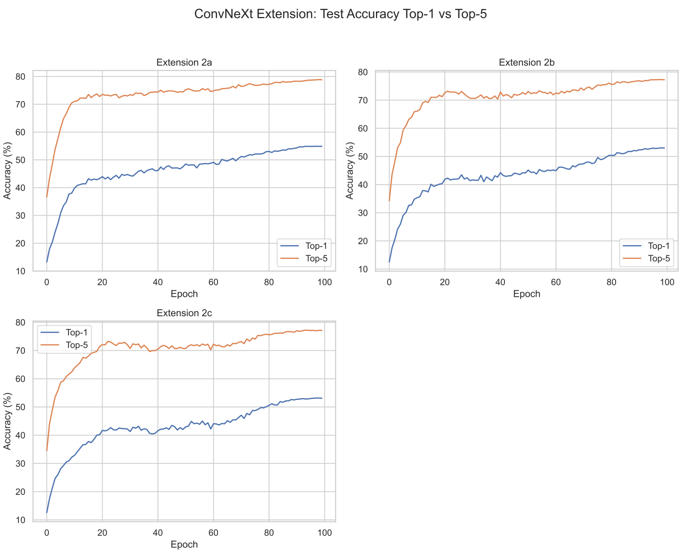

Figure 6: Top 1 vs Top 5 for BN, ReLU, Restore 1x1 conv respectively.

The experimental results demonstrate that for every architectural component tested, reverting to the traditional design yielded a performance improvement rather than a degradation. Specifically, restoring the 1×1 convolutional bottleneck and replacing GELU with ReLU resulted in accuracy gains of +0.29% and +0.15%, respectively. Most notably, substituting LayerNorm with BatchNorm led to a substantial +2.02% increase in Top-1 accuracy. This stands in stark contrast to the original ImageNet findings, where these same modifications caused accuracy drops ranging from -0.3% to -0.6%.

These results highlight a significant scale-dependency in the ConvNeXt design space. The original architecture was optimized for high-resolution images (224×224) and heavy data augmentation (Mixup, CutMix, etc.). Our findings suggest that in "low-regime" settings, which is characterized by low resolution (32×32) and the absence of augmentation, modern choices like LayerNorm and GELU may introduce unnecessary complexity or stabilization overhead that hinders learning.

The observation that traditional designs (BN, ReLU, standard bottlenecks) outperform modern "Transformer-like" designs on CIFAR-100 implies that the ConvNeXt modernization process is deeply tied to the scale of the dataset and the intensity of the training recipe. While LayerNorm and inverted bottlenecks are foundational for high-capacity models on large-scale datasets, they do not appear to be universal improvements for convolutional networks. Instead, our results advocate for a more nuanced application of these design choices, tailored specifically to the resolution and data volume of the target task.

4.4 Extension 3: Medical Imaging Transfer

The results for cross-domain transfer learning on the HAM10000 dataset reveal significant performance disparities between the tested architectures (Table 4). ConvNeXt-Tiny achieved the highest performance, reaching a Top-1 accuracy of 90.47% and a macro F1-score of 0.835, substantially outperforming both the ResNet-50 and Swin-Tiny baselines.

Table 4: Transfer learning results on HAM10000 (7-class skin lesion classification). All models fine-tuned from CIFAR-100 weights.

Model	Top-1 (%)	Macro F1
ConvNeXt-Tiny	**90.47**	**0.835**
ResNet-50	79.35	0.563
Swin-Tiny (Default)	69.68	0.228
Swin-Tiny (Optimised)*	79.80	0.620

* Trained with slower LR, 60 epochs, and stronger regularisation.

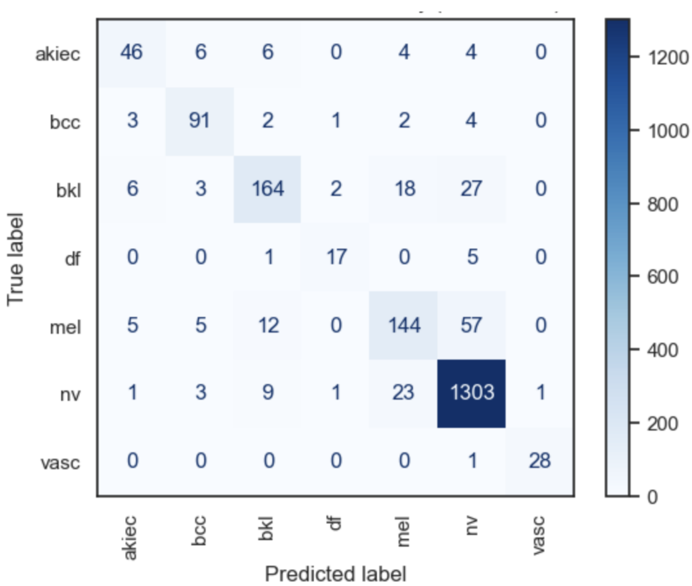

Figure 7: Confusion matrix for ConvNeXt-Tiny.

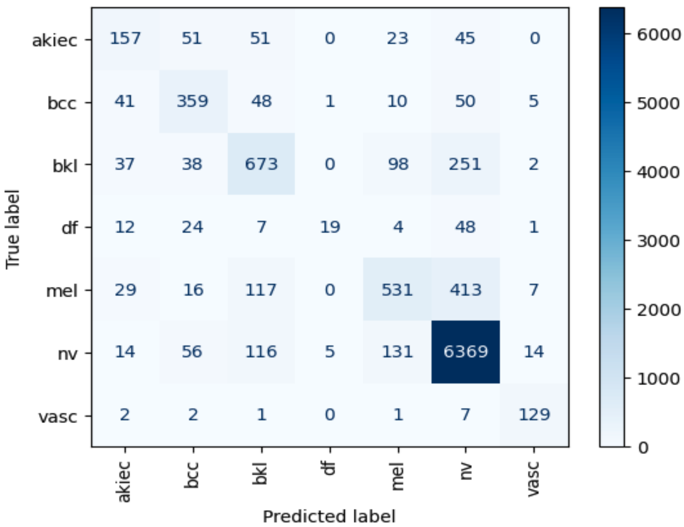

Figure 8: Confusion matrix for ResNet-50.

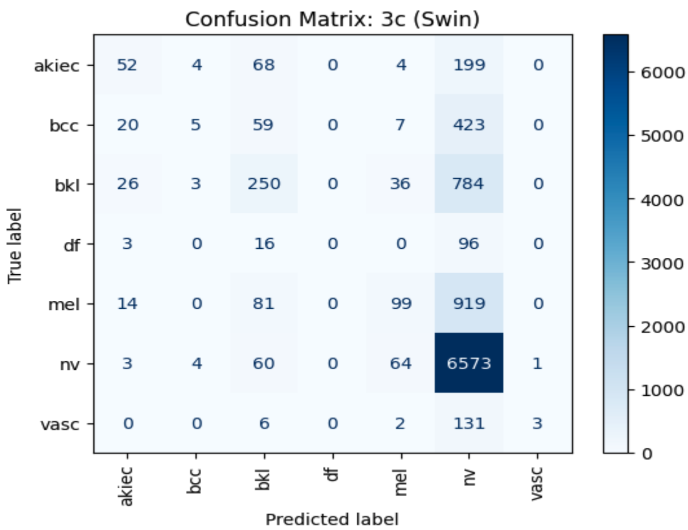

Figure 9: Confusion matrix for Swin-Tiny (Default).

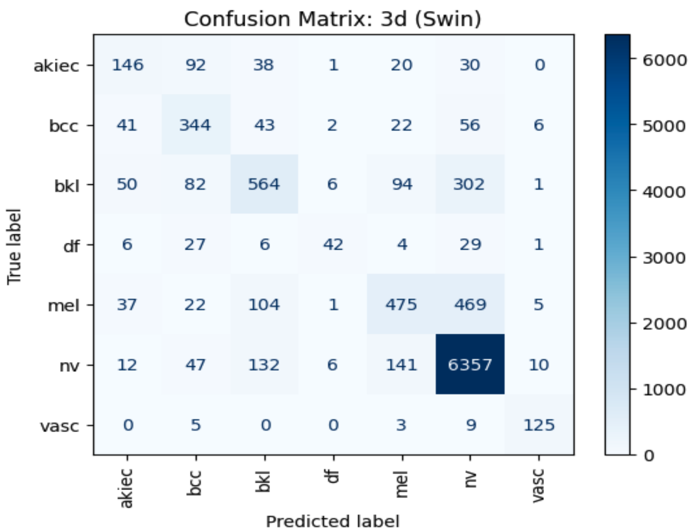

Figure 10: Confusion matrix for Swin-Tiny (Optimised).

The default Swin-Tiny configuration initially struggled with the HAM10000 dataset, exhibiting a low macro F1-score (0.228) and a decent Top-1 Accuracy score that suggest a failure to generalise across imbalanced clinical classes (the model is good at predicting one dominant class). However, after applying a more conservative training recipe, including a slower learning rate, extended training to 60 epochs, and stronger regularisation, Swin-Tiny improved significantly to 79.80% accuracy and macro F1-score to 0.620.

Despite these optimisations, ConvNeXt-Tiny maintained a lead of over 10 percentage points in both accuracy and F1-score. The confusion matrix (Fig. 7) also proves that ConvNeXt-Tiny treats every class as equally important regardless of how many samples it has, a high macro F1-score (0.835) means the model is performing well across the classes.

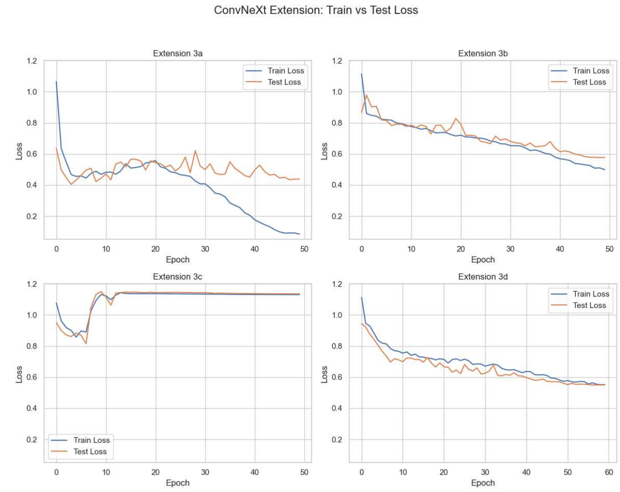

Figure 11: Train vs Test Loss for ConvNeXt-Tiny, ResNet-50, Swin-Tiny (Default), Swin-Tiny (Optimised).

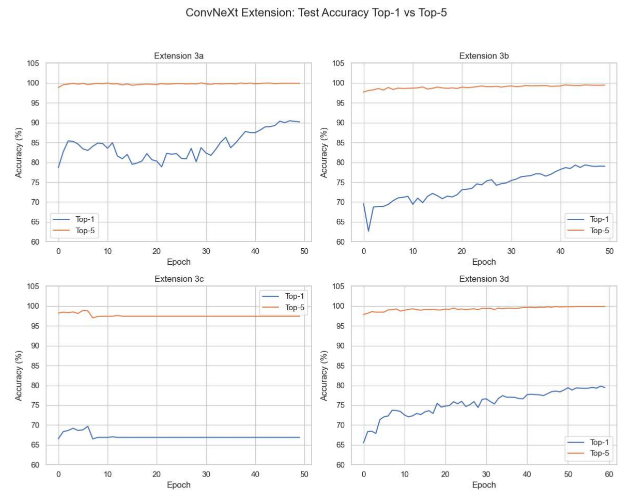

Figure 12: Top 1 vs Top 5 for ConvNeXt-Tiny, ResNet-50, Swin-Tiny (Default), Swin-Tiny (Optimised).

As shown in the training dynamics (Fig. 12), ConvNeXt-Tiny demonstrated superior convergence and stability. While ResNet-50 and Swin-Tiny showed consistent loss reduction, ConvNeXt-Tiny's ability to maintain high Top-5 accuracy (near 100%) early in training suggests its modern micro-design choices, such as LayerNorm and larger kernels, provide a more robust feature representation for the fine-grained textures required in skin lesion classification. This confirms that ConvNeXt's architectural benefits generalise effectively beyond ordinary images to specialised visual domains (medical imaging).

5. Discussion

Our CIFAR-100 reproduction achieved 55.1% Top-1 accuracy, only marginally outperforming ResNet-50 (52.89%) by 2.21 percentage points, which is substantially smaller than the 5–10% gains reported on ImageNet (Liu et al., 2022). However, the main advantage lies in improved generalisation behaviour: ConvNeXt consistently maintains a narrower train-test loss gap across all augmentation settings, indicating better calibration and reduced overfitting due to the heavy regularisation recipe (Mixup, CutMix, drop path). This discrepancy between modest accuracy gains and improved generalisation metrics suggests that ConvNeXt's design benefits may manifest differently depending on dataset characteristics. ImageNet's 1.28 million high-resolution (224×224) diverse images provide ample opportunity for sophisticated augmentations and architectural modernisations to demonstrate their full potential, whereas CIFAR-100's limited scale (50,000 images) and low resolution (32×32) constrain how much performance can be extracted from these techniques.

Extension 1 reveals that geometric augmentations (+3.43%) substantially outperform mixture methods on CIFAR-100, contradicting ImageNet findings where Mixup/CutMix dominate. Adding mixture augmentations atop geometric transforms actually *decreases* accuracy by 1.94%, suggesting soft-labelling interferes with learning fine-grained boundaries at low resolution.

Extension 2's architectural verification produces the opposite result from ImageNet: reverting to BatchNorm (+2.02%), ReLU (+0.15%), and standard bottlenecks (+0.29%) all *improve* accuracy. This inversion exposes a critical scale-dependency: LayerNorm and GELU, designed for high-resolution Transformer training (Ba et al., 2016; Hendrycks & Gimpel, 2016), introduce misaligned inductive biases when applied to small, spatially homogeneous 32×32 patches where BatchNorm's translation invariance is more appropriate.

Extension 3 demonstrates ConvNeXt's strongest advantage, which is its 90.47% accuracy on HAM10000 medical imaging outperforming ResNet-50 and Swin-Tiny by >10%. The 7×7 depthwise convolutions effectively capture local texture patterns (lesion boundaries, pigment gradients) critical for dermoscopy, while Swin-Tiny's lack of built-in spatial locality bias requires extensive hyperparameter tuning to achieve comparable performance.

5.1 Limitations

Extensions 1 and 2 faced implementation challenges requiring multiple reruns. Initially, RandAugment remained enabled in the baseline configuration, artificially inflating accuracy and obscuring marginal contributions of other augmentation families. After disabling RandAugment, ColorJitter and RandomErasing were still active by default in the data pipeline, necessitating additional debugging to achieve a true no-augmentation baseline. Each 100-epoch training run required ∼3 hours on Kaggle T4 GPU, making reruns slow and limiting extensive hyperparameter exploration. CIFAR-100's small scale (50,000 images) and 32×32 resolution limit generalisability to ImageNet-scale benchmarks, and computational constraints restricted study to ConvNeXt-Tiny (28.6M parameters) — larger variants may exhibit different scaling behaviour.

Extension 3's HAM10000 dataset is heavily imbalanced ("nv" class >60%), causing models to develop bias toward predicting "nv" more frequently, which is dangerous in medical diagnosis if rarer, serious conditions (e.g., melanoma) are missed. Clinical deployment would require additional calibration techniques such as class-balanced sampling to ensure reliable detection of minority classes (Cui et al., 2019). A key finding was that Swin-Tiny (Vision Transformer) does not naturally understand that neighbouring pixels are related, unlike CNNs (ResNet/ConvNeXt), requiring slower learning rates and longer warmup (60 epochs vs ConvNeXt's 50) to stabilise during medical transfer. HAM10000 images were downsampled to 224×224 for Kaggle compatibility, removing vital textural details (pigment networks, vascular patterns) that could represent clinically significant diagnostic features, limiting accuracy compared to real-world applications using higher-resolution inputs.

6. Conclusion

This project reproduced ConvNeXt-Tiny on CIFAR-100 (55.1% accuracy, +2.21% over ResNet-50) and revealed critical scale-dependencies through three extensions. Extension 1 showed geometric augmentations outperform mixture methods on CIFAR-100 (+3.43% vs −1.94%), contradicting ImageNet trends. Extension 2 found traditional designs (BatchNorm, ReLU, standard bottlenecks) *improve* accuracy on CIFAR-100 (+2.02%, +0.15%, +0.29%), opposing ImageNet ablations and exposing resolution-dependent inductive bias mismatches. Extension 3 demonstrated strong medical imaging transfer (90.47% on HAM10000, >10% above baselines), confirming ConvNeXt's advantages for texture-rich domains.

ConvNeXt's thesis that architectural modernisation rivals attention mechanisms holds conditionally: it excels on large-scale, high-resolution tasks and specialised domains (medical imaging) but underperforms traditional CNNs on small, low-resolution datasets where Transformer-inspired choices introduce unnecessary complexity. Success stems from co-design of architecture, augmentation, and data scale rather than architectural innovation alone. Practitioners must tailor design choices to dataset characteristics rather than universally applying ImageNet best practices.

Future work should evaluate larger ConvNeXt variants on CIFAR-100, extend medical imaging experiments to histopathology and radiology, explore hybrid CNN-attention architectures (Dai et al., 2021), and develop adaptive augmentation schedules and resolution-aware normalisation to reconcile ImageNet-CIFAR discrepancies.

References

Ba, J. L., Kiros, J. R., & Hinton, G. E. (2016). Layer normalization. *arXiv preprint arXiv:1607.06450*. https://arxiv.org/abs/1607.06450

Cui, Y., Jia, M., Lin, T.-Y., Song, Y., & Belongie, S. (2019). Class-balanced loss based on effective number of samples. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 9268–9277. https://doi.org/10.1109/CVPR.2019.00949

Dai, Z., Liu, H., Le, Q. V., & Tan, M. (2021). Coatnet: Marrying convolution and attention for all data sizes. *Advances in Neural Information Processing Systems (NeurIPS)*, *34*, 3965–3977. https://proceedings.neurips.cc/paper/2021/hash/20568692db622456cc42a2e853ca21f8-Abstract.html

Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., & Houlsby, N. (2021). An image is worth 16x16 words: Transformers for image recognition at scale. *Proceedings of the International Conference on Learning Representations (ICLR)*. https://doi.org/10.48550/arXiv.2010.11929

He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 770–778. https://doi.org/10.1109/CVPR.2016.90

Hendrycks, D., & Gimpel, K. (2016). Gaussian error linear units (gelus). *arXiv preprint arXiv:1606.08415*. https://arxiv.org/abs/1606.08415

Liu, Z., Lin, Y., Cao, Y., Hu, H., Wei, Y., Zhang, Z., Lin, S., & Guo, B. (2021). Swin transformer: Hierarchical vision transformer using shifted windows. *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 10012–10022. https://doi.org/10.1109/ICCV48922.2021.00986

Liu, Z., Mao, H., Wu, C.-Y., Feichtenhofer, C., Darrell, T., & Xie, S. (2022). A ConvNet for the 2020s. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 11976–11986. https://doi.org/10.1109/CVPR52688.2022.01167

Sandler, M., Howard, A., Zhu, M., Zhmoginov, A., & Chen, L.-C. (2018). MobileNetV2: Inverted residuals and linear bottlenecks. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 4510–4520. https://doi.org/10.1109/CVPR.2018.00474

Tschandl, P., Rosendahl, C., & Kittler, H. (2018). The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions. *Scientific Data*, *5*, 180161. https://doi.org/10.1038/sdata.2018.161

Xie, S., Girshick, R., Dollár, P., Tu, Z., & He, K. (2017). Aggregated residual transformations for deep neural networks. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 5987–5995. https://doi.org/10.1109/CVPR.2017.634

Zhang, H., Cisse, M., Dauphin, Y. N., & Lopez-Paz, D. (2018). Mixup: Beyond empirical risk minimization. *International Conference on Learning Representations (ICLR)*. https://openreview.net/forum?id=r1Ddp1-Rb